

SOFTWARE DESIGN DOCUMENT for BRAINSTORMING APPLICATION

Issue No: 1.0
Issue Date: Dec 2025

Prepared by
Selen Erdoğan
Barış Eren Gezici
Ömer Faruk Koç
Muhammed Bilal Türk
Mehmet Yücel Kaplaner

	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---

TABLE OF CONTENTS

1	Introduction	3
1.1	Purpose of the system.....	3
1.2	Design goals.....	3
1.3	Definitions, acronyms, and abbreviations.....	3
1.4	References.....	3
1.5	Overview.....	3
2	Current software architecture	3
3	Proposed software architecture	3
3.1	Overview.....	3
3.2	Subsystem decomposition.....	3
3.3	Hardware/software mapping.....	3
3.4	Persistent data management.....	4
3.5	Access control and security.....	4
3.6	Global software control.....	4
3.7	Boundary conditions.....	4
4	Subsystem services	4
5	Glossary of Terms	4
6	Traceability	4

	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---

1 Introduction

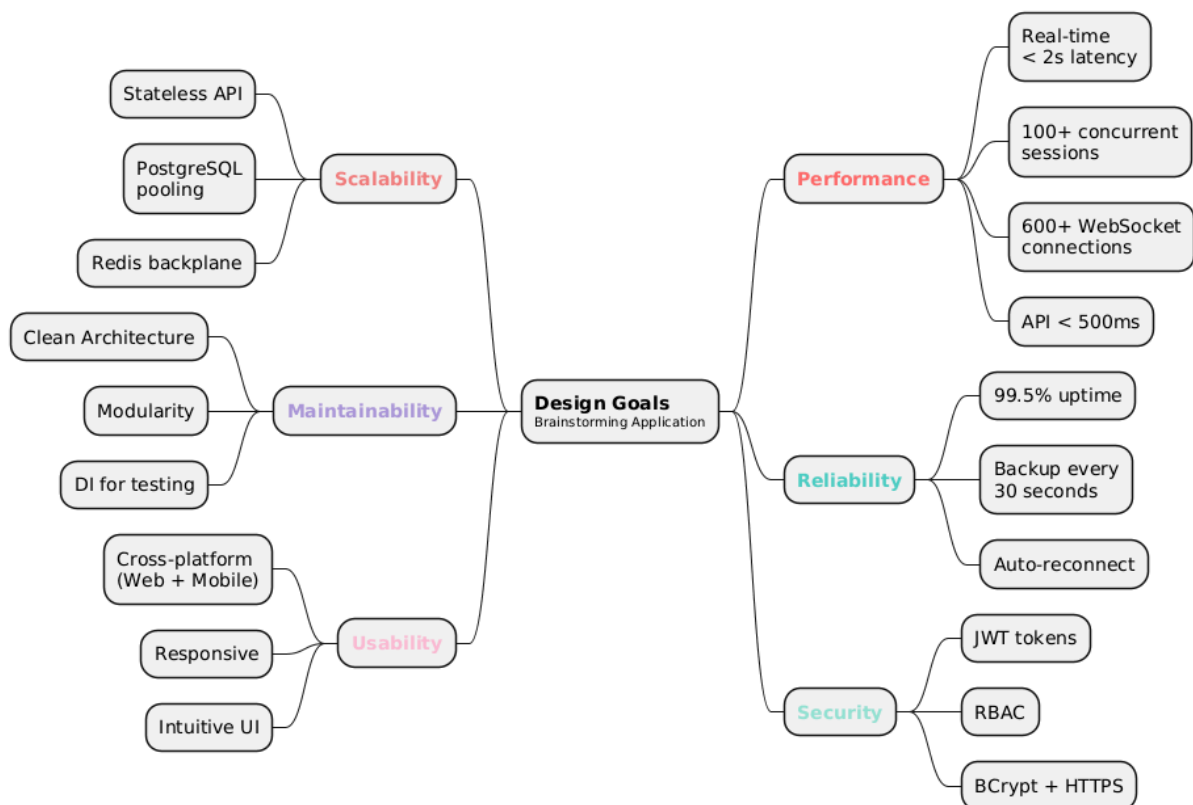
1.1 Purpose of the system

The Brainstorming Platform is a web-based and mobile application designed to facilitate structured brainstorming sessions using the proven 6-3-5 Method. This method enables six participants to generate three ideas each over five rounds, resulting in 90+ ideas in approximately 25-30 minutes.

The system aims to:

- Digitalize the traditional paper-based 6-3-5 brainstorming method
- Enable remote collaboration for distributed teams
- Provide AI-powered assistance for idea generation and session insights
- Track and document brainstorming sessions with comprehensive audit trails
- Facilitate event management for multiple concurrent brainstorming activities

1.2 Design goals



	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---

Fig.1.1

Figure 1.1 illustrates the six primary design goals that guide the architectural decisions for the Brainstorming Application. As shown in the mind map, these goals are organized into Performance (real-time responsiveness with sub-2-second latency and support for 100+ concurrent sessions), Reliability (99.5% uptime with automated backups every 30 seconds), Security (JWT-based authentication with role-based access control), Scalability (stateless API design), Maintainability (Clean Architecture principles with modular design and dependency injection for testability), and Usability (cross-platform support with responsive design for web and mobile). Each category, represented by distinct color-coded branches in the diagram, addresses specific quality attributes that work in concert to create a robust, scalable, and user-friendly collaborative platform, with particular emphasis on maintainability as the foundation for long-term system evolution through separation of concerns, independent subsystems with clear interfaces, and comprehensive unit testing capabilities.

1.3 Definitions, acronyms, and abbreviations

6-3-5 Method , A structured brainstorming technique where 6 participants develop 3 ideas in 5 rounds

API ,Application Programming Interface

CORS, Cross-Origin Resource Sharing

DTO, Data Transfer Object

EF Core, Entity Framework Core - ORM for .NET

JWT, JSON Web Token - authentication standard

ORM, Object-Relational Mapping

RBAC, Role-Based Access Control

REST, Representational State Transfer

SignalR, Real-time communication library for .NET

SPA, Single Page Application

WebSocket, Full-duplex communication protocol

	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---

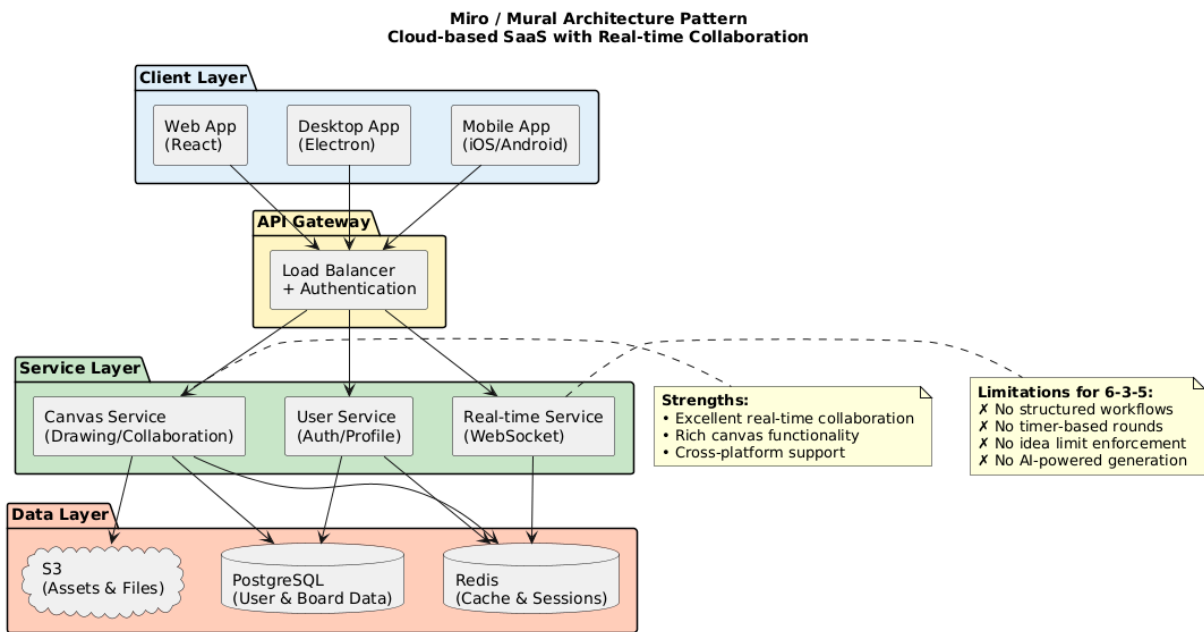
1.4 Overview

- This Software Design Document is organized into six main sections:
- Section 2 describes current architectures of similar systems and their limitations
 - Section 3 presents the proposed software architecture including subsystem decomposition, deployment, data management, security, control flow, and boundary conditions
 - Section 4 details the services provided by each subsystem
 - Section 5 provides a glossary of domain-specific terms
 - Section 6 establishes traceability between this SDD and the RAD

2 Current software architecture

Since there is no existing system being replaced, this section surveys current architectures of similar brainstorming and collaboration tools.

2.1 Miro / Mural (Digital Whiteboards)



	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---

Fig 2 Miro and Mural

Miro and Mural employ a cloud-based SaaS architecture where client applications (Web built with React, Desktop, and Mobile apps) communicate through an API Gateway that handles load balancing and authentication, which then distributes requests to three main microservices: Canvas Service for drawing and collaboration features, User Service for authentication and profile management, and Real-time Service using WebSocket for live synchronization, with the backend supported by PostgreSQL for persistent storage of user and board data, Redis for caching and session management, and S3 for asset storage, though this architecture lacks structured brainstorming workflows like the 6-3-5 method, timer-based round management, participant idea limit enforcement, and AI-powered idea generation capabilities.

2.2 Google Docs / Microsoft 365 (Collaborative Documents)

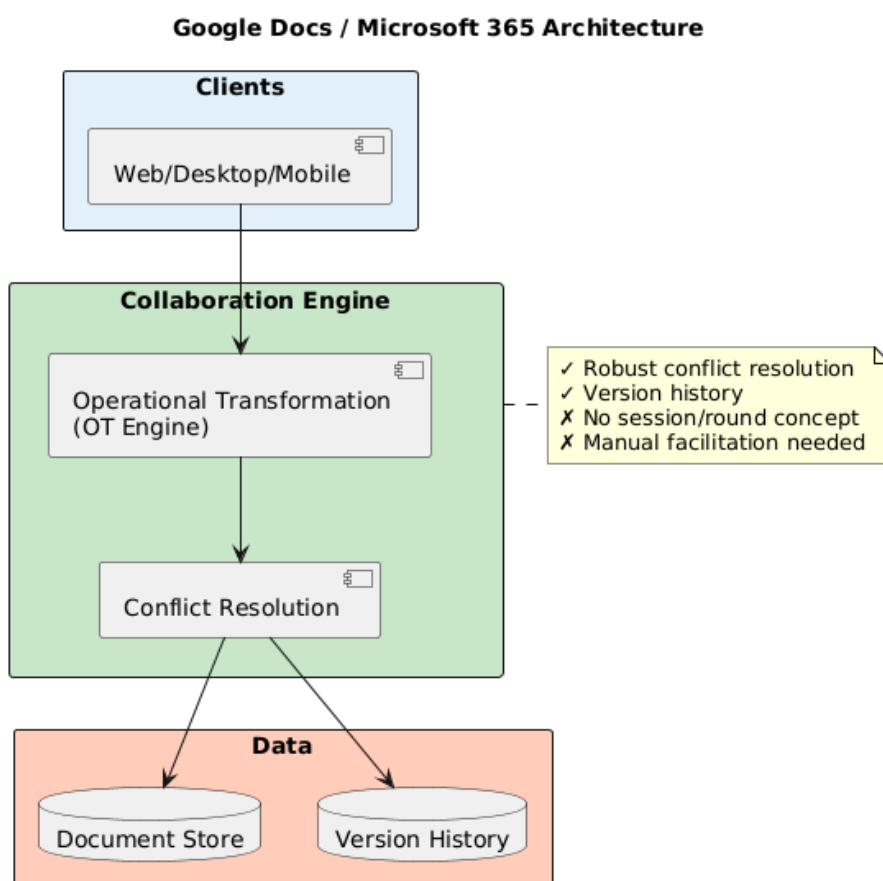


Fig 3 -Google Docs and Microsoft 365

Google Docs and Microsoft 365 utilize Operational Transformation (OT) based architecture where multiple client applications (Web, Desktop, Mobile) connect to a

	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---

central collaboration engine that manages real-time concurrent editing through conflict resolution algorithms and maintains comprehensive version history in document storage, providing robust multi-user editing and wide accessibility, but this document-centric approach lacks structured brainstorming frameworks like session and round concepts, requires manual facilitation for timed activities, and offers no real-time participant synchronization specifically designed for collaborative ideation workflows such as the 6-3-5 method.

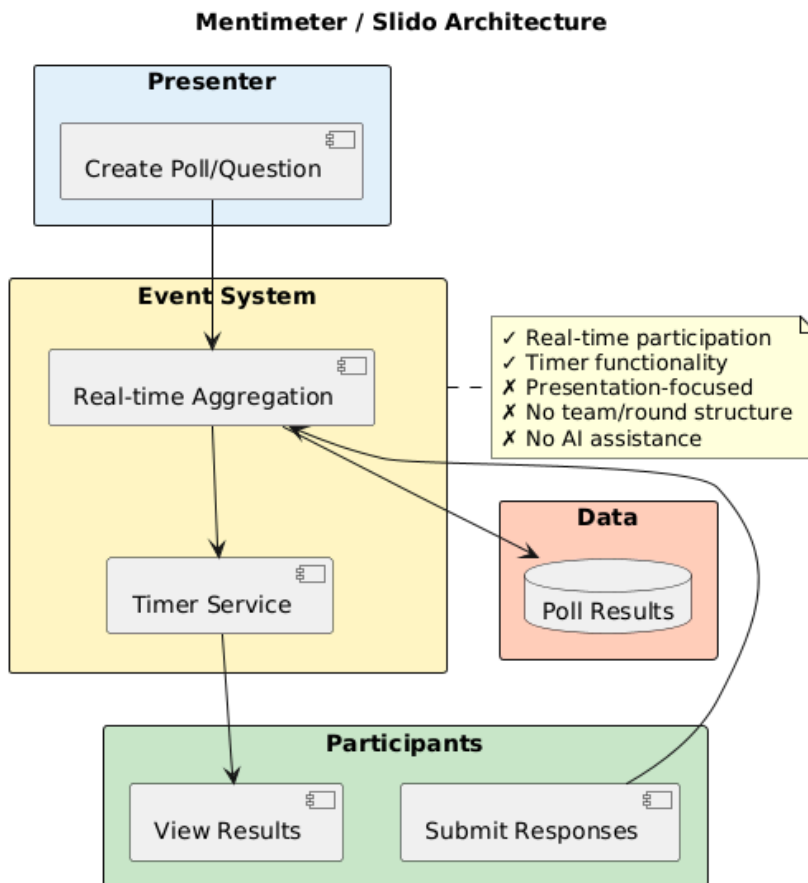


Fig.4 Mentimeter and Slido

Mentimeter and Slido implement an event-driven polling system where a presenter creates questions or polls that are distributed to participants who submit responses through simple web interfaces, with a real-time aggregation engine collecting and displaying results immediately while a timer service manages activity duration and stores poll results in a centralized database, excelling at real-time audience participation and timed interactions for presentations, but this architecture is

	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	--	---------------------------------------

fundamentally designed for one-way presenter-to-audience engagement rather than collaborative ideation, lacking team and round structure necessary for iterative brainstorming, offering limited idea building capabilities where participants cannot develop others' contributions, and providing no AI-powered assistance for idea generation or enhancement.

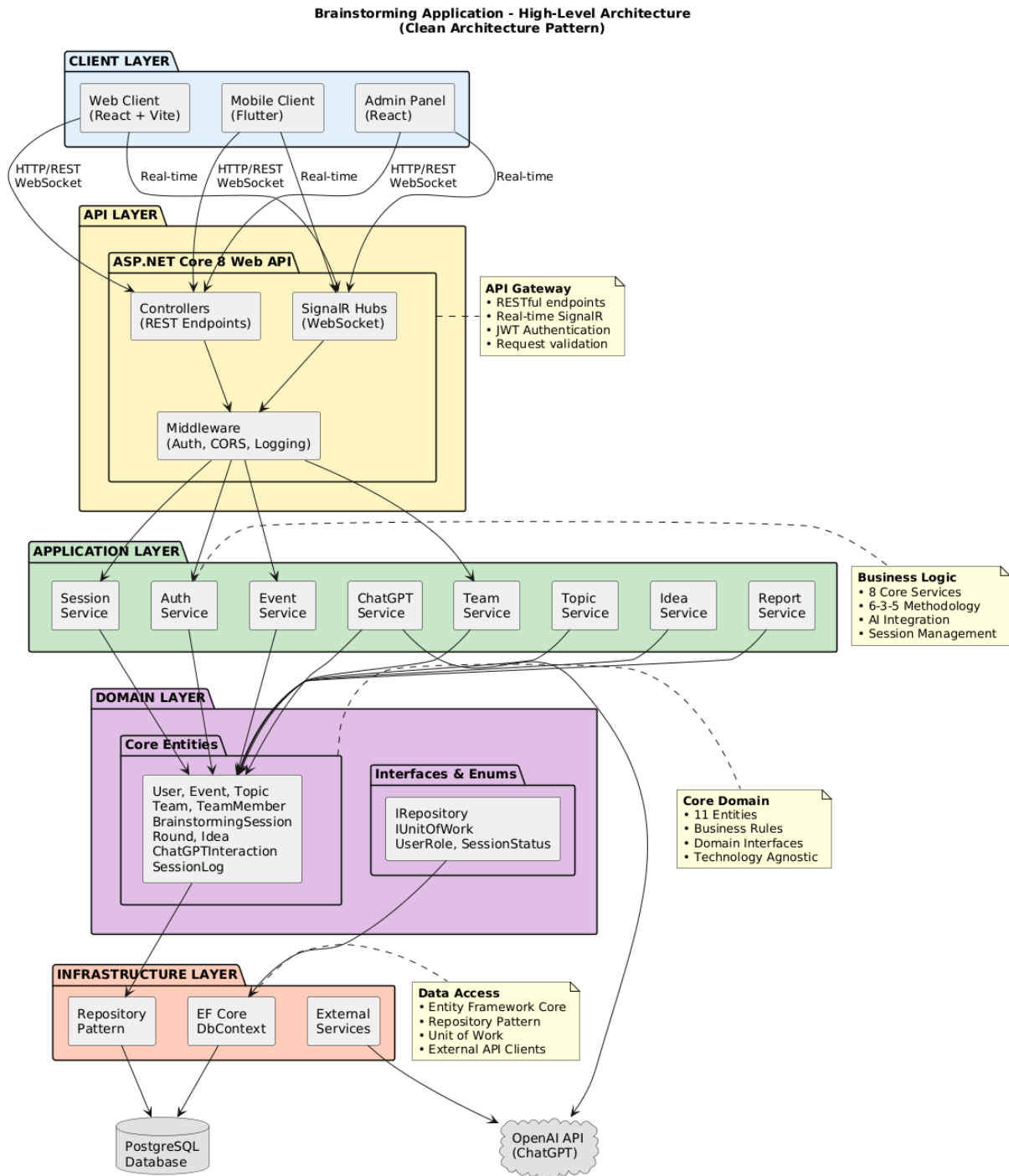
3 Proposed software architecture

The third section documents the system design model of the new system.

3.1 Overview

The Brainstorming Application follows a Clean Architecture pattern with clear separation of concerns. The system is designed as a distributed application with a RESTful API backend, real-time communication layer, and cross-platform frontend clients.

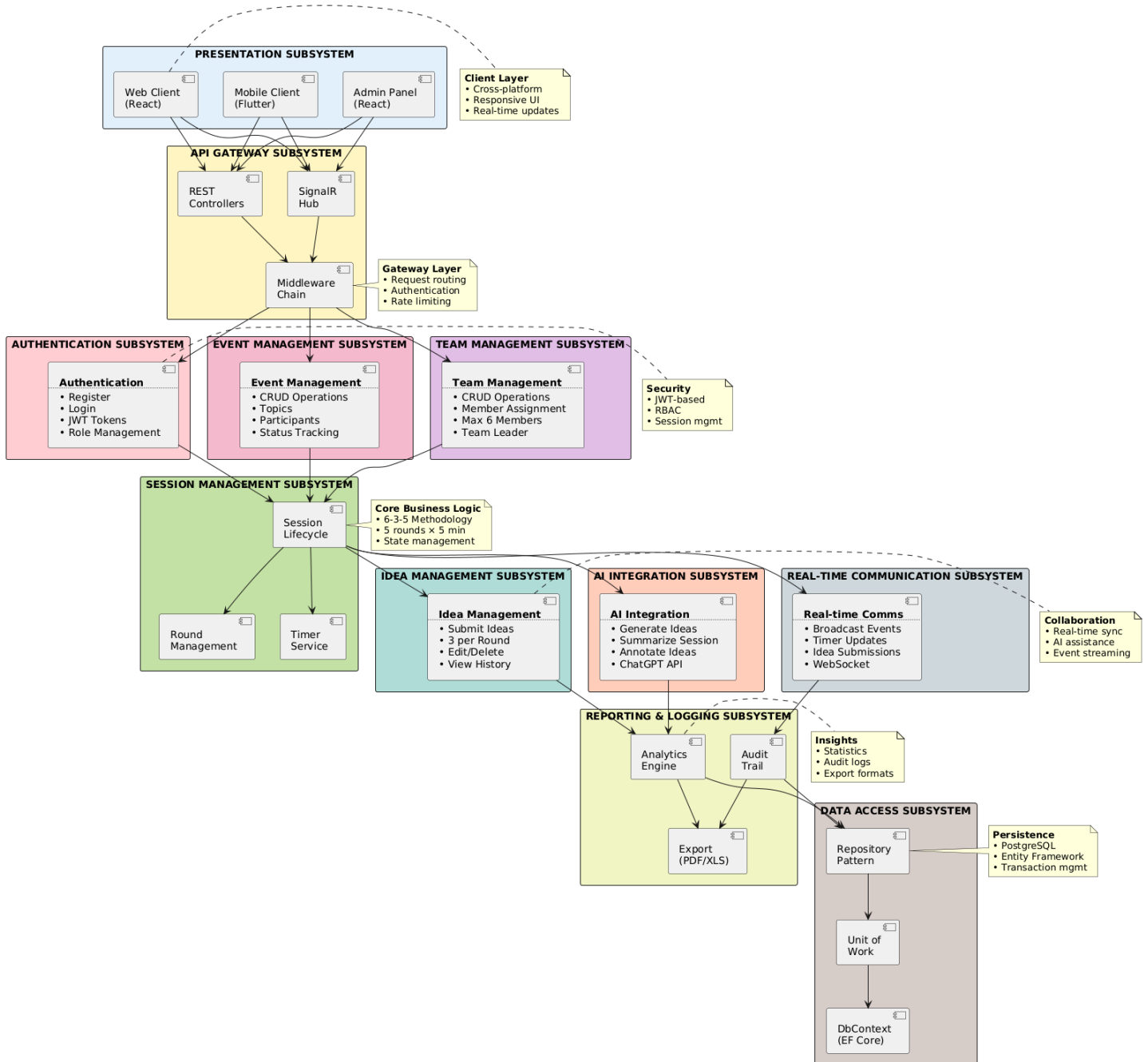
	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---

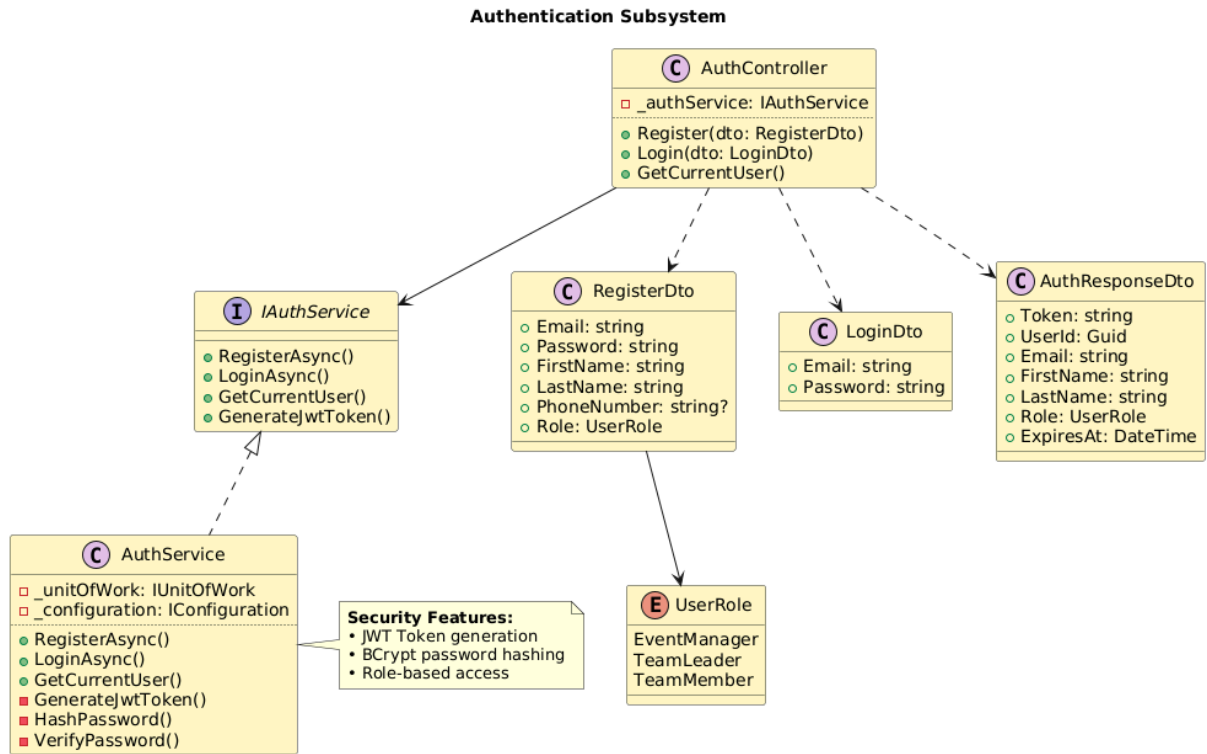


3.2 Subsystem decomposition

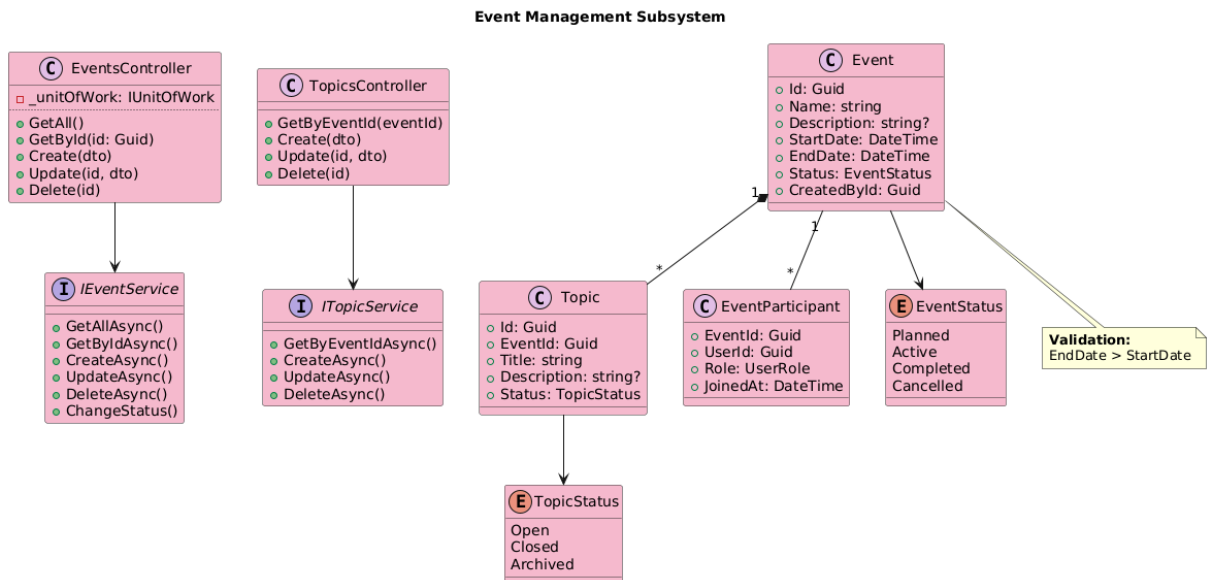
Subsystem decomposition describes the decomposition into subsystems and the responsibilities of each. This is the main product of system design.

Brainstorming Application - Subsystem Decomposition

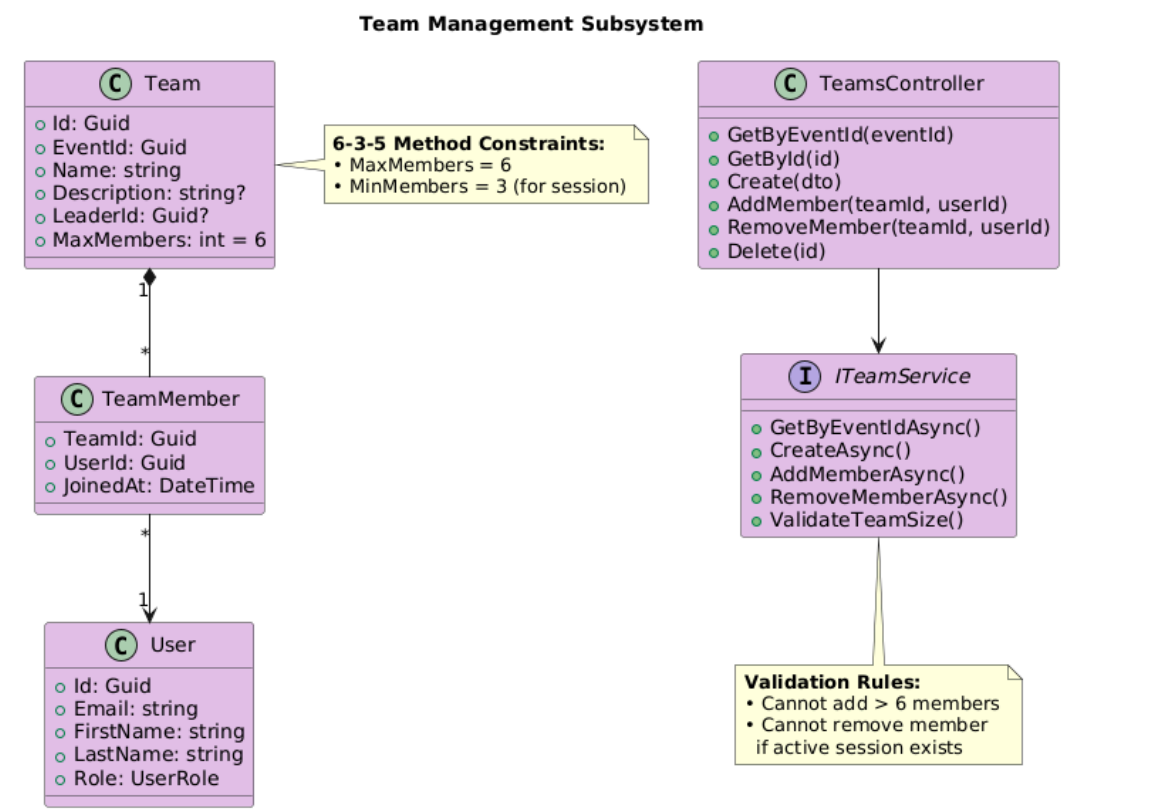




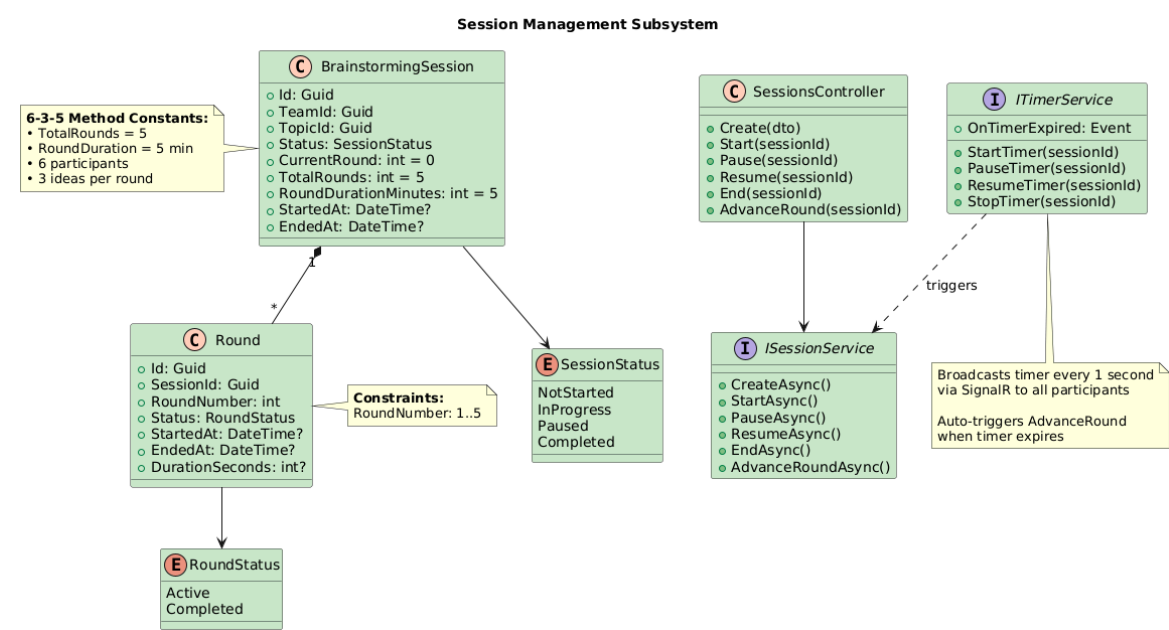
Authentication Subsystem



Event Management Subsystem

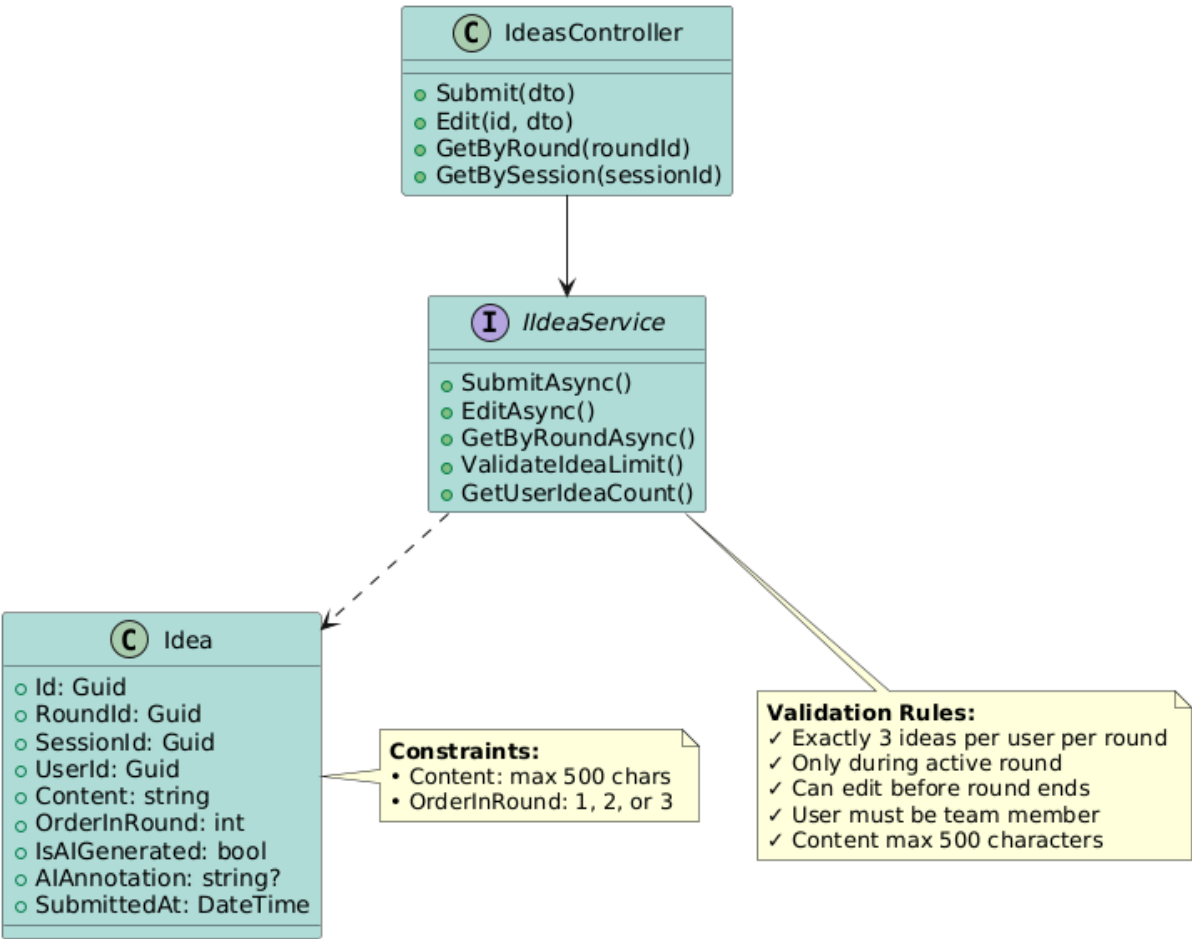


Team Management Subsystem

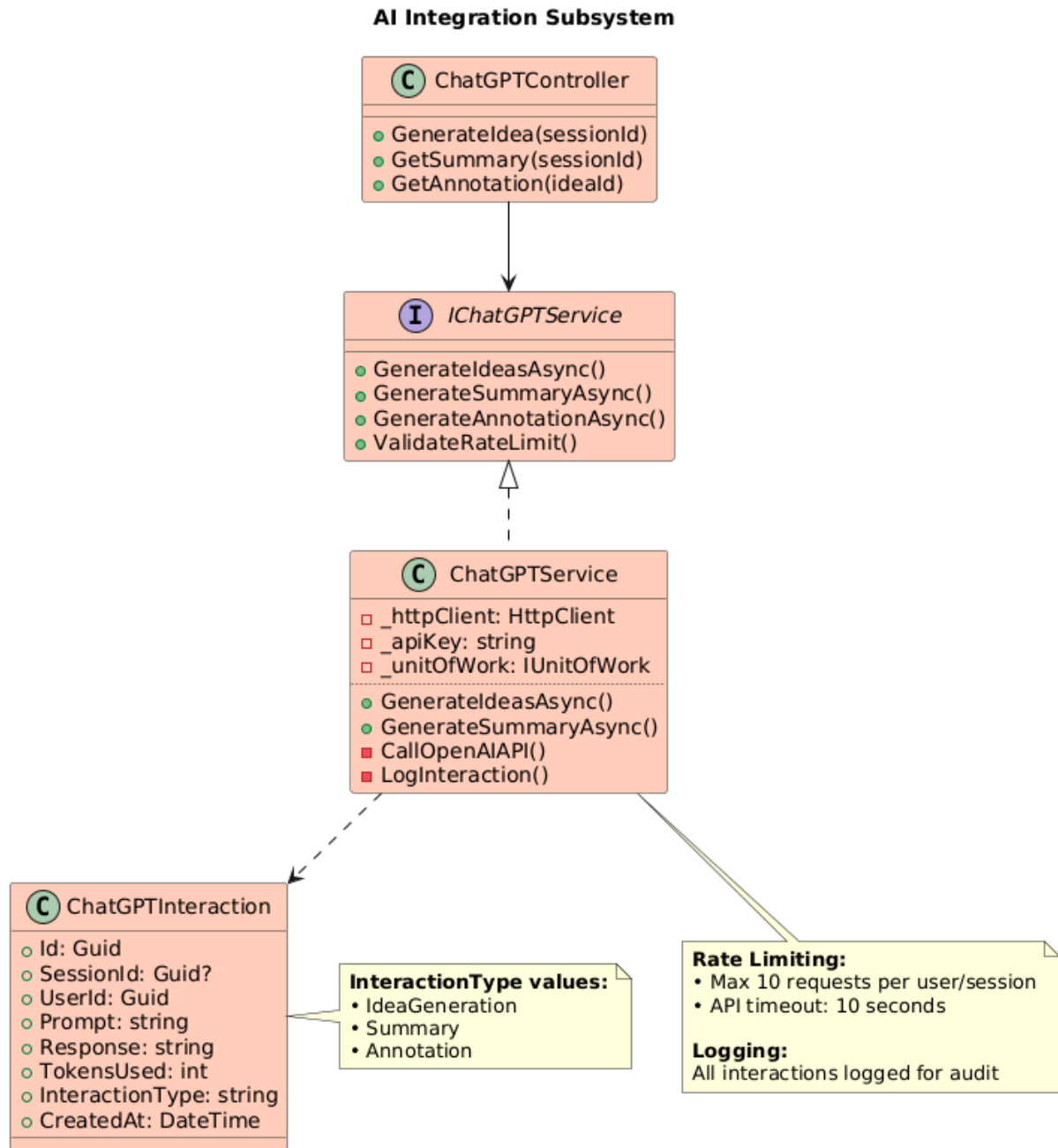


Session Management Subsystem

Idea Management Subsystem



Idea Management Subsystem



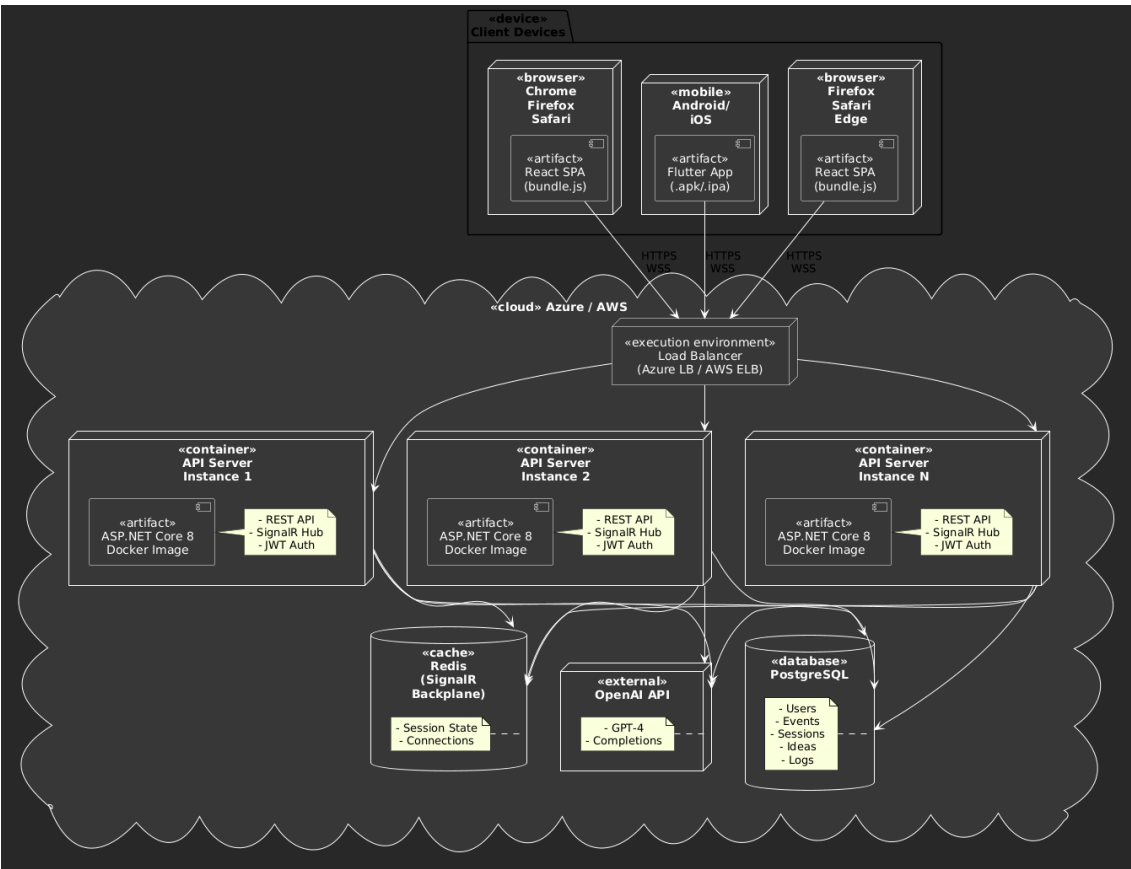
AI Integration Subsystem

	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---

Real-time Communication Subsystem

3.3 Hardware/software mapping

3.3.1 Deployment Diagram



3.3.2 Node Specifications

Node	Description	Technology	Scaling
Client Browser	Web application	React 18, TypeScript, Vite	N/A
Client Mobile	Mobile application	Flutter 3.x, Dart	N/A
Load Balancer	Traffic distribution	Azure LB / AWS ELB / Nginx	N/A
API Server	Backend services	.NET 8, Docker	Horizontal (N instances)
Database	Persistent storage	PostgreSQL 15	Vertical + Read replicas
Cache	SignalR backplane, session state	Redis 7.x	Cluster mode
External API	AI services	OpenAI GPT-4 API	N/A (managed service)

3.3.3 Communication Protocols

	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	--	---------------------------------------

Connection	Protocol	Port	Purpose
Client → Load Balancer	HTTPS	443	REST API requests
Client → API Server	WSS	443	SignalR real-time
API Server → Database	TCP	5432	PostgreSQL connection
API Server → Redis	TCP	6379	Cache & SignalR backplane
API Server → OpenAI	HTTPS	443	AI API calls

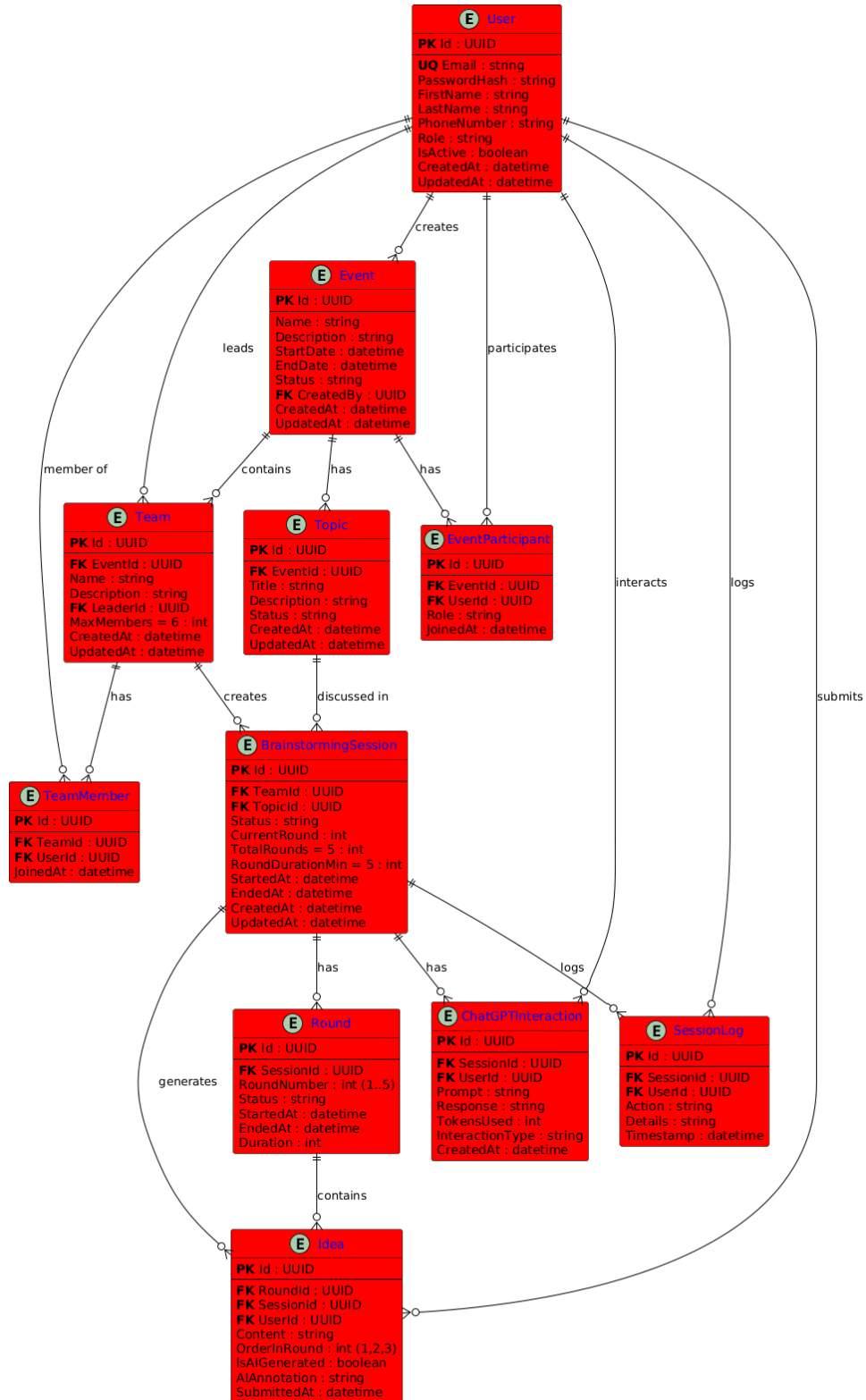
3.4 Persistent data management

3.4.1 Database Schema



SOFTWARE DESIGN DOCUMENT for Brainstorming Application

Issue No: 1.0
Issue Date: Dec 2025

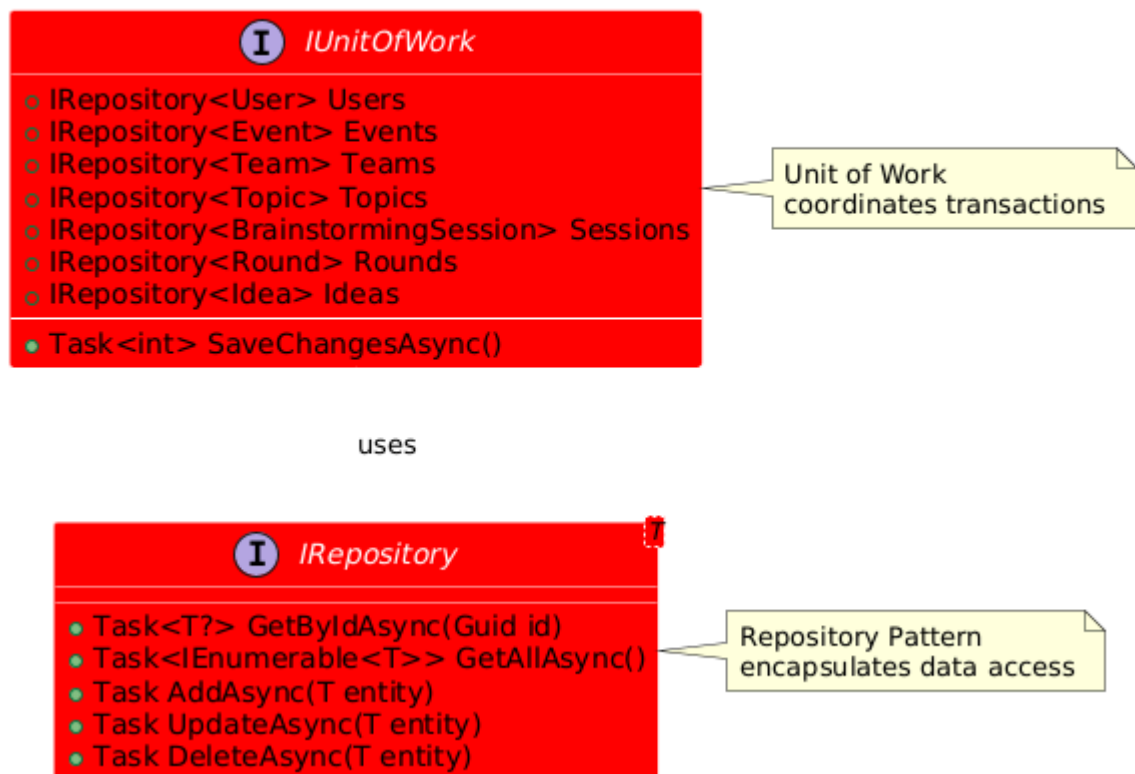


	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---

3.4.2 Database Technology

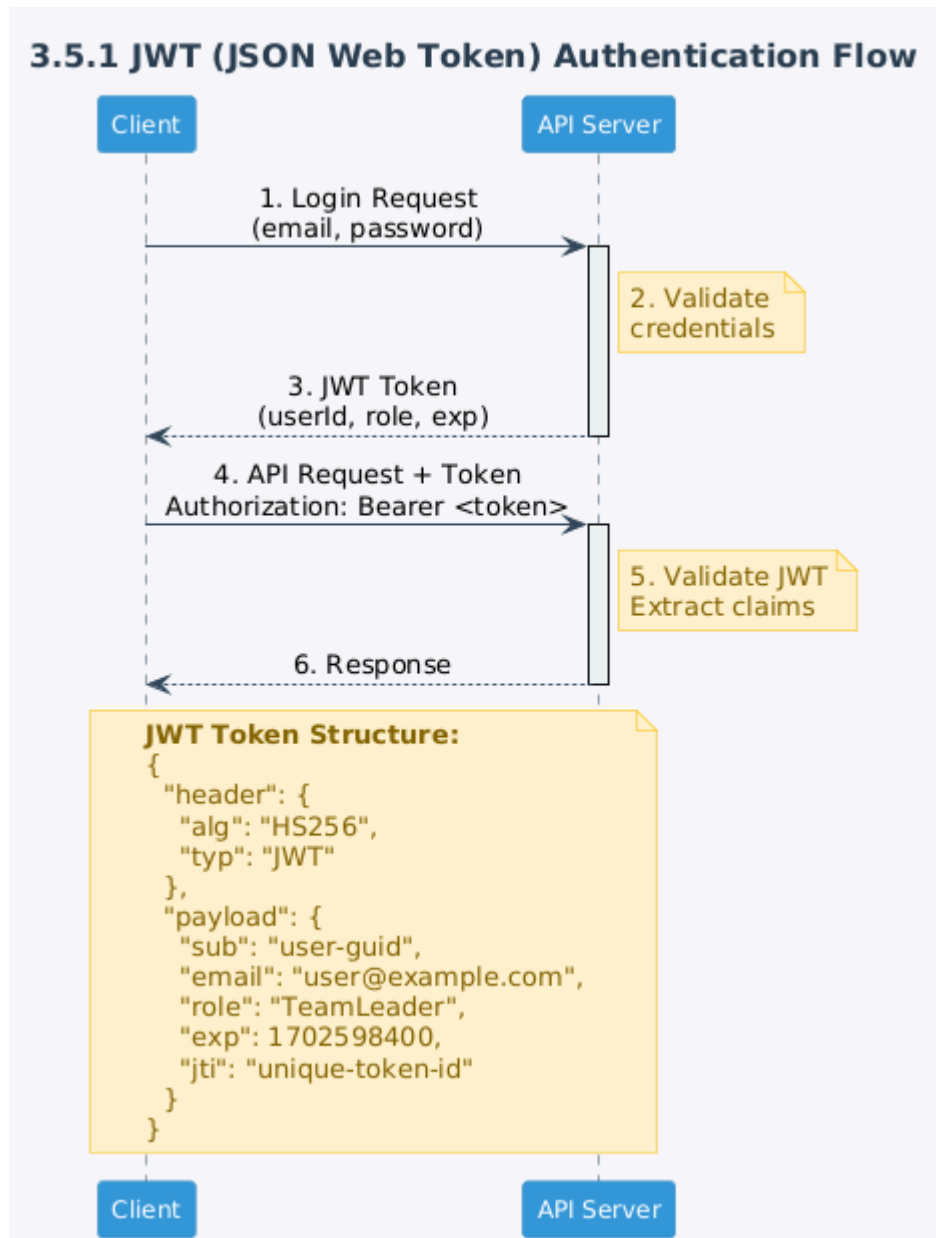
Aspect	Choice	Rationale
RDBMS	PostgreSQL 15	Open-source, ACID compliant, excellent JSON support
ORM	Entity Framework Core 8	.NET native, Code-First migrations, LINQ support
Connection Pooling	Npgsql built-in	Efficient connection management
Migrations	EF Core Migrations	Version-controlled schema changes

3.4.3 Data Access Strategy



	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---

3.5 Access control and security



	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---

3.5.2 Access Control Matrix

Resource / Action	EventManager	TeamLeader	TeamMember
User Management			
Register	✓	✓	✓
Login	✓	✓	✓
View own profile	✓	✓	✓
Update own profile	✓	✓	✓
Event Management			
Create event	✓	✗	✗
Update event	✓	✗	✗
Delete event	✓	✗	✗
View events	✓	✓	✓
Add participants	✓	✗	✗
Topic Management			
Create topic	✓	✗	✗
Update topic	✓	✗	✗
Delete topic	✓	✗	✗
View topics	✓	✓	✓
Team Management			
Create team	✓	✓	✗
Add members	✓	✓ (own team)	✗
Remove members	✓	✓ (own team)	✗
View team	✓	✓	✓ (own team)
Session Management			
Create session	✗	✓	✗
Start session	✗	✓ (own team)	✗
Pause/Resume session	✗	✓ (own team)	✗
End session	✗	✓ (own team)	✗
View session	✓	✓	✓ (own team)
Idea Management			
Submit idea	✗	✓	✓
Edit own idea	✗	✓	✓
View ideas	✓	✓	✓ (own session)
AI Features			
Generate AI idea	✗	✓	✓
Get session summary	✓	✓	✗
Reports			
View session analytics	✓	✓ (own team)	✗
Export reports	✓	✓ (own team)	✗
View audit logs	✓	✓ (own team)	✗

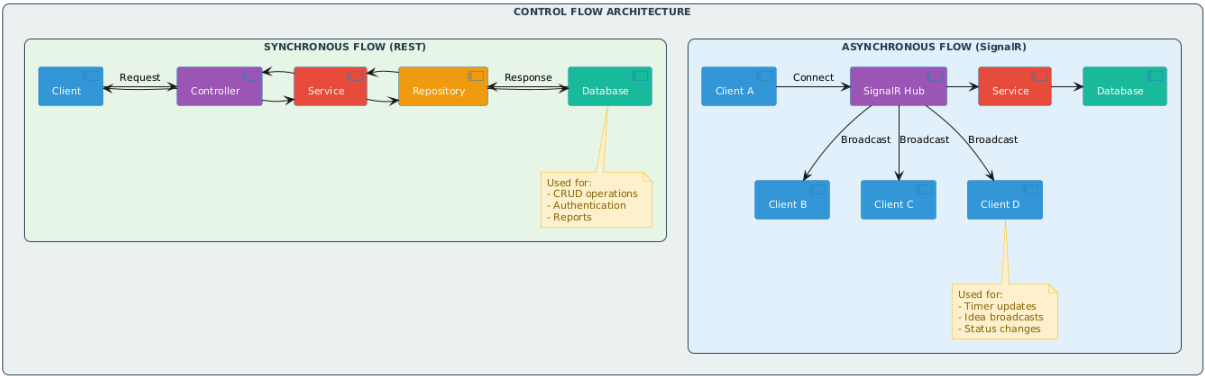
3.5.3 Security Measures

Security Aspect	Implementation
Password Storage	BCrypt hashing with salt (work factor 10)
Token Security	HMAC-SHA256 signed JWT, 24-hour expiry
Transport Security	HTTPS/TLS 1.3 enforced
CORS	Whitelist of allowed origins
Input Validation	Server-side validation on all inputs
SQL Injection	Parameterized queries via EF Core
XSS Prevention	Output encoding, Content-Security-Policy
Rate Limiting	API rate limits, AI request limits per session

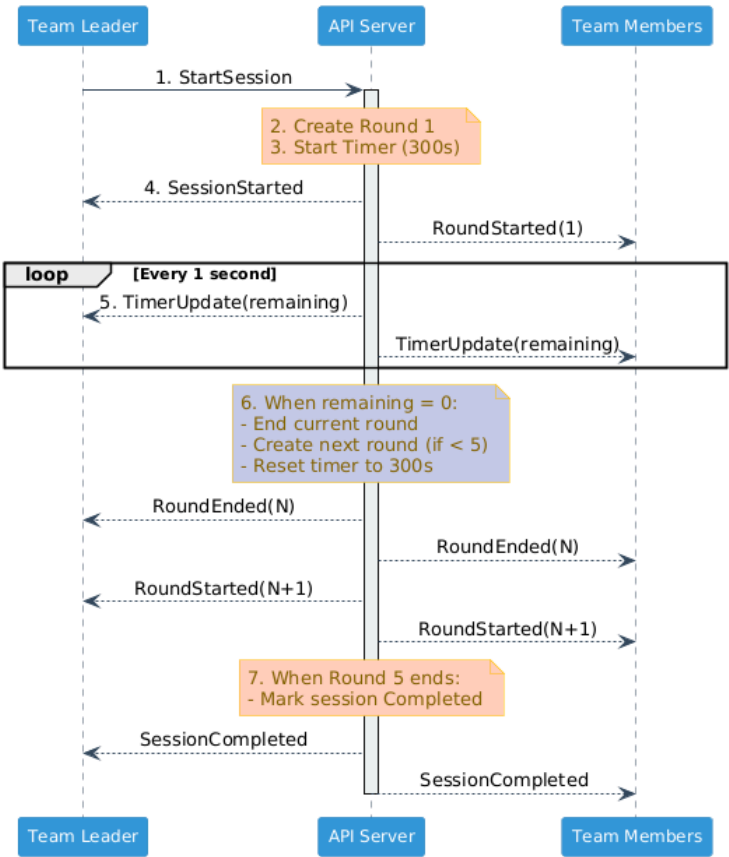
	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---

3.6 Global software control

3.6.1 Control Flow Architecture



3.6.2 Session Timer Control Flow



	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---

3.6.3 Concurrency Handling

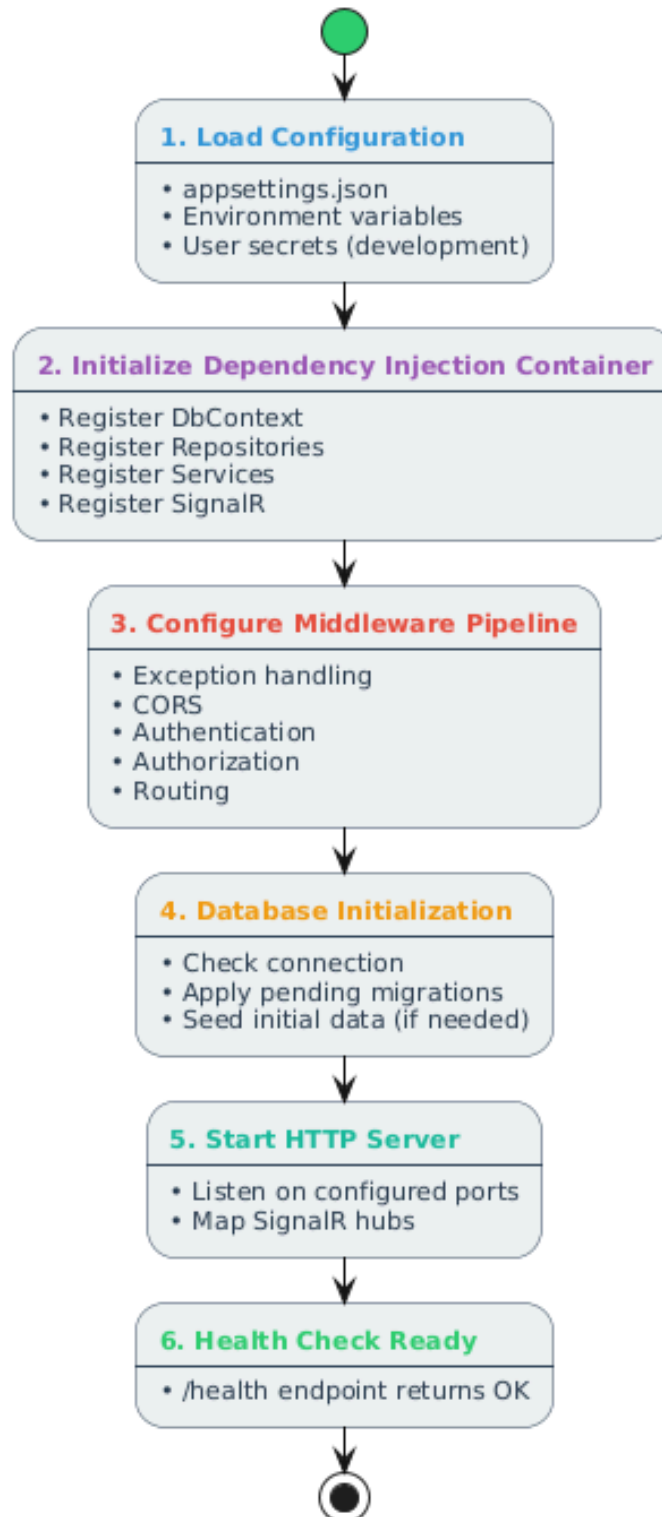
Scenario	Strategy
Simultaneous idea submissions	Optimistic concurrency with timestamp
Timer synchronization	Server-authoritative, broadcast to all
Session state changes	Pessimistic locking on session record
Database transactions	Unit of Work pattern with SaveChanges
SignalR scaling	Redis backplane for multi-instance

	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---

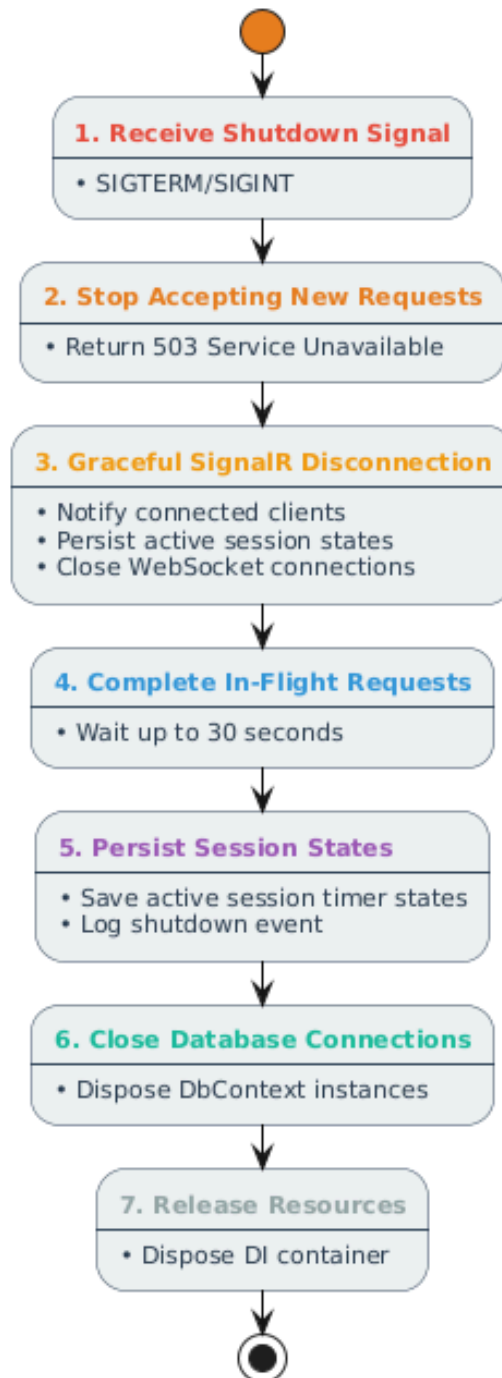
3.7 Boundary conditions

	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---

3.7.1 System Startup Sequence



3.7.2 System Shutdown Sequence



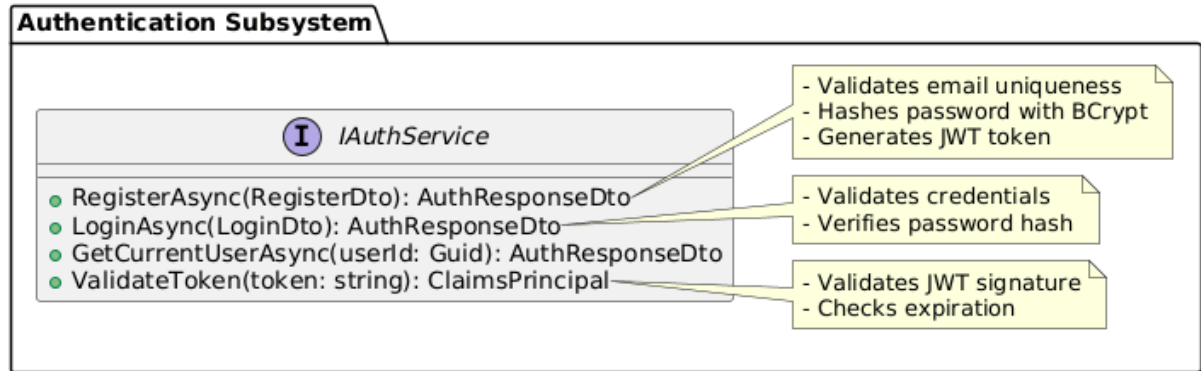
3.7.3 Error Handling

Error Type	Handling Strategy
Validation errors	Return 400 Bad Request with details
Authentication failure	Return 401 Unauthorized
Authorization failure	Return 403 Forbidden
Resource not found	Return 404 Not Found
Business rule violation	Return 409 Conflict with message
Database errors	Log, return 500 Internal Server Error
External service timeout	Retry with exponential backoff, fallback
SignalR disconnection	Auto-reconnect with state recovery

3.7.4 Failure Recovery

Failure Scenario	Recovery Strategy
Server crash during session	Session state persisted every 30s; recover on restart
Database connection lost	Connection pool retry; circuit breaker pattern
Redis unavailable	Fallback to in-memory (single instance)
OpenAI API timeout	Return graceful error; allow manual idea entry
Client disconnection	SignalR auto-reconnect; rejoin session group

4 Subsystem services



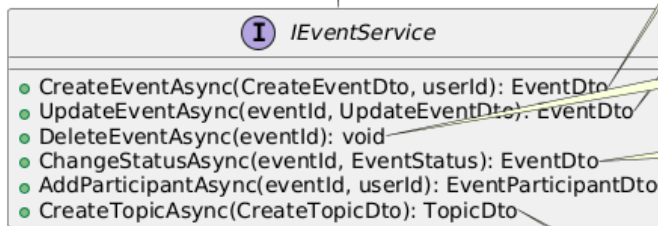


SOFTWARE DESIGN DOCUMENT for Brainstorming Application

Issue No: 1.0
Issue Date: Dec 2025

Event Management Subsystem

CreateEventAsync İş Akışı



- Validates date range (end > start)
- Creates event with Planned status
- Sets creator reference

- Validates event exists
- Updates event properties

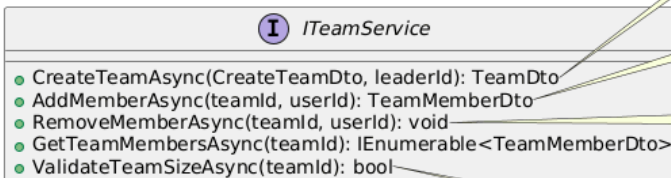
- Validates no associated sessions exist
- Deletes event and related data

- Validates status transition
- Updates event status

- Validates user and event exist
- Prevents duplicate participation
- Sends notification to user

- Validates event exists and is accessible
- Creates topic with Open status

Team Management Subsystem



- Validates event exists
- Sets max members to 6 (6-3-5 constraint)
- Assigns team leader

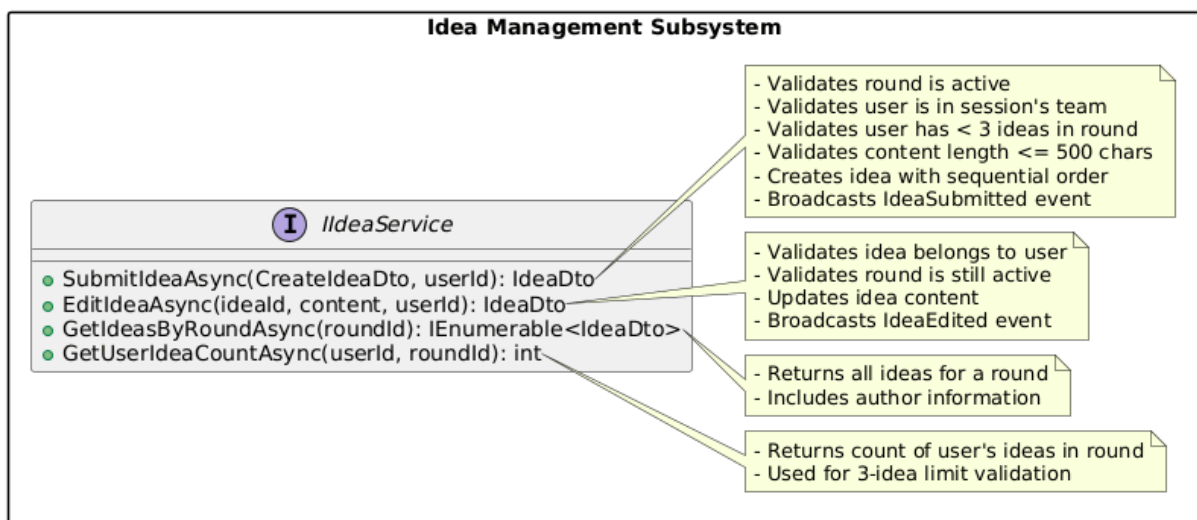
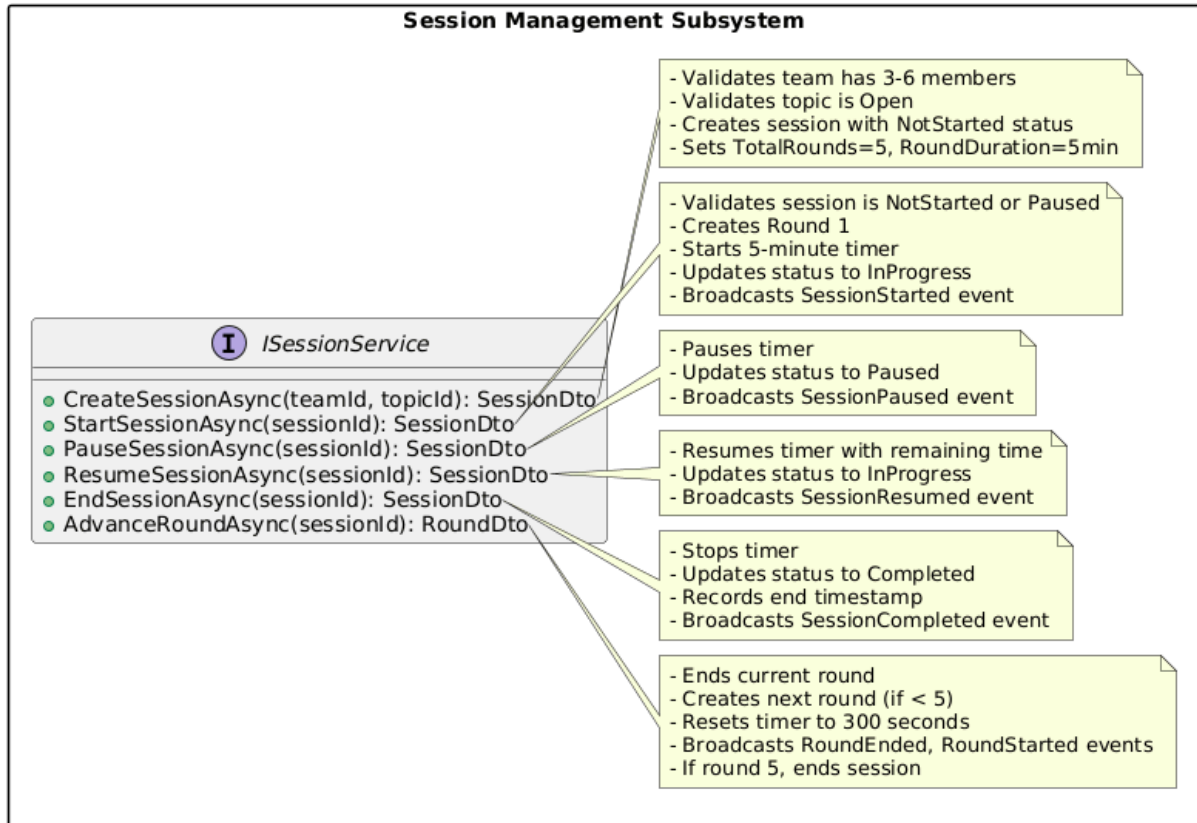
- Validates team has space (< 6 members)
- Validates user not already in team
- Creates team membership

- Validates member has no active sessions
- Removes team membership

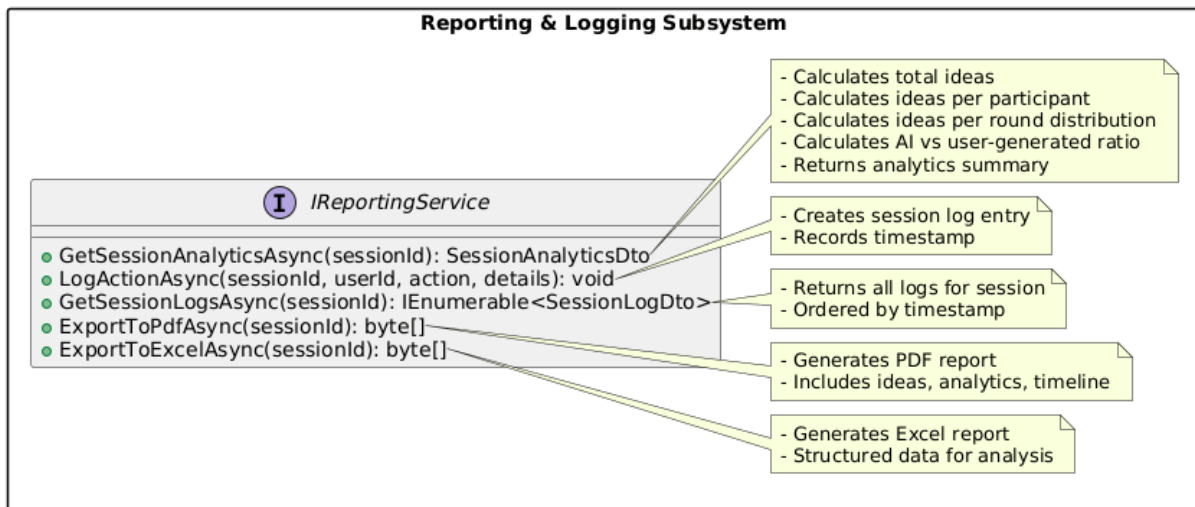
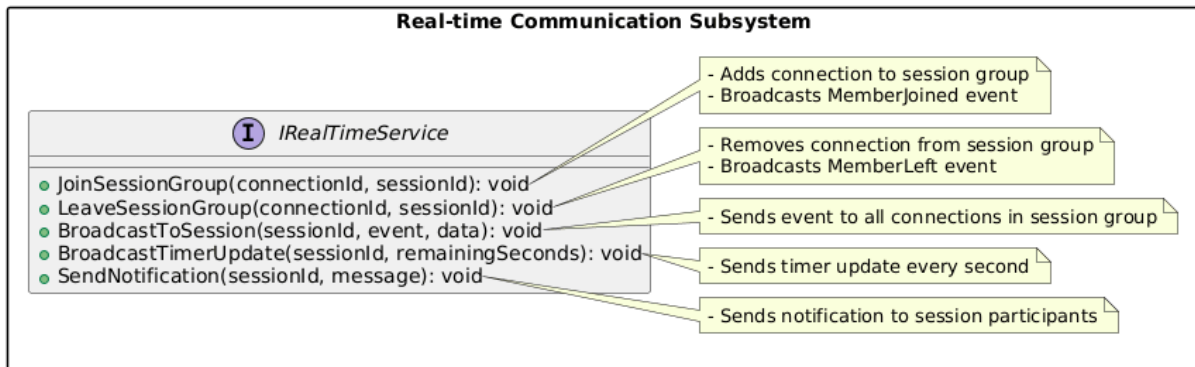
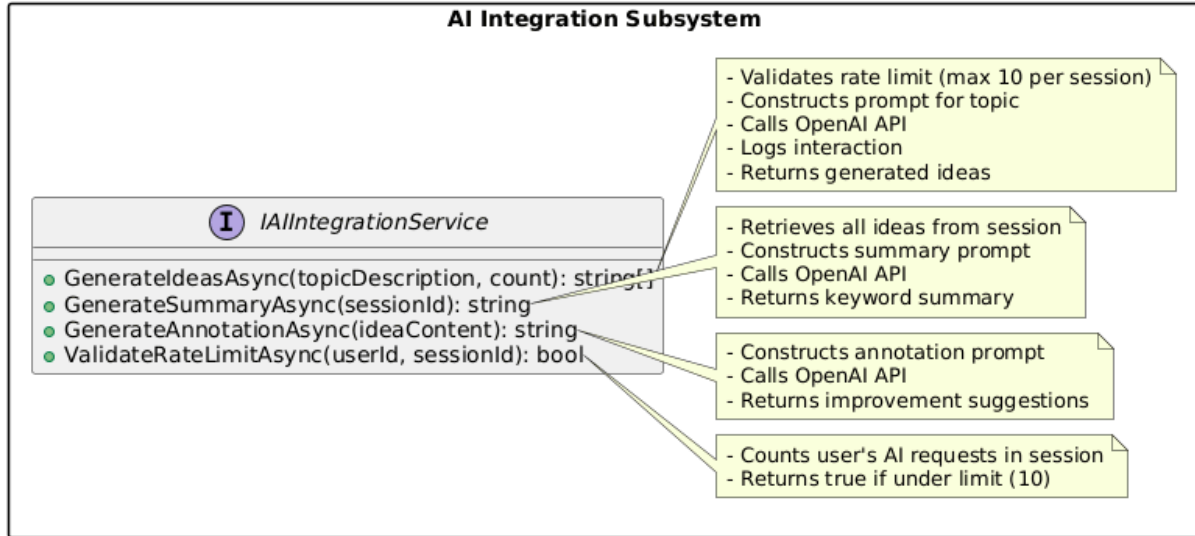
- Returns all team members with user details

- Returns true if team has 3-6 members
- Used before session creation

	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---



	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---



	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---

5 Glossary of Terms

Term	Definition
6-3-5 Method	<i>A structured brainstorming technique where 6 participants each write 3 ideas in 5 rounds of 5 minutes each</i>
Access Control Matrix	<i>A table specifying which roles can perform which operations on which resources</i>
API Gateway	<i>Entry point for all client requests, handling routing, authentication, and load balancing</i>
Backplane	<i>Infrastructure allowing multiple server instances to share state (e.g., Redis for SignalR)</i>
Boundary Conditions	<i>System behavior during startup, shutdown, and error scenarios</i>
Clean Architecture	<i>Software design pattern separating concerns into layers (Domain, Application, Infrastructure, Presentation)</i>
CRUD	<i>Create, Read, Update, Delete - basic data operations</i>

	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---

DTO	<i>Data Transfer Object - object used to transfer data between layers</i>
Entity	<i>Domain object with identity persisted in database</i>
Hub	<i>SignalR server-side component managing real-time connections</i>
JWT	<i>JSON Web Token - compact, URL-safe means of representing claims between parties</i>
Middleware	<i>Software components that handle cross-cutting concerns in request pipeline</i>
ORM	<i>Object-Relational Mapping - technique for converting data between type systems</i>
Repository Pattern	<i>Abstraction over data access providing collection-like interface for entities</i>
SignalR	<i>Library for adding real-time web functionality using WebSockets</i>
Subsystem	<i>A grouping of related functionality with well-defined interfaces</i>
Unit of Work	<i>Pattern maintaining list of objects affected by transaction and coordinating persistence</i>

	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---

WebSocket	<i>Protocol providing full-duplex communication over single TCP connection</i>
------------------	--

6 Traceability

6.1 Requirements to Subsystems Traceability Matrix

Requirement ID	Requirement Description	Mapped Subsystem
FR-1 to FR-10	User registration, login, profile, RBAC	Authentication Subsystem
FR-11 to FR-20	Event CRUD, status management	Event Management Subsystem
FR-21 to FR-27	Topic CRUD, assignment	Event Management Subsystem
FR-28 to FR-37	Team CRUD, member management	Team Management Subsystem
FR-38 to FR-43	Event participation	Event Management Subsystem
FR-44 to FR-60	Session lifecycle management	Session Management Subsystem

	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---

FR-61 to FR-67	Round management, timer	Session Management Subsystem
FR-68 to FR-81	Idea submission, validation	Idea Management Subsystem
FR-82 to FR-91	ChatGPT integration	AI Integration Subsystem
FR-92 to FR-102	Real-time communication	Real-time Communication Subsystem
FR-103 to FR-111	Session logging, audit	Reporting & Logging Subsystem
FR-112 to FR-118	Analytics, export	Reporting & Logging Subsystem

6.2 Non-Functional Requirements to Design Decisions

NFR ID	Requirement	Design Decision
NFR-1 to NFR-5	Usability	React SPA with responsive design, real-time feedback

	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---

NFR-6 to NFR-10	Reliability	Session state persistence, auto-reconnect, backups
NFR-11 to NFR-16	Performance	Horizontal scaling, Redis cache, async operations
NFR-17 to NFR-18	Supportability	Correlation IDs, health endpoints
NFR-19 to NFR-21	Interface	OpenAPI/Swagger, OAuth 2.0 support
NFR-22 to NFR-24	Packaging	Docker containers, CI/CD pipeline
NFR-25 to NFR-27	Legal	Consent checkbox, MIT/Apache licensed dependencies

6.3 Use Cases to Subsystems Traceability

Use Case	Primary Subsystem	Supporting Subsystems
----------	-------------------	-----------------------

	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---

UC-1: Register/Login	Authentication	Data Access
UC-2: Manage Events	Event Management	Authentication, Data Access
UC-3: Manage Topics	Event Management	Authentication, Data Access
UC-4: Manage Teams	Team Management	Authentication, Data Access
UC-5: Event Participation	Event Management	Authentication, Data Access
UC-6: Manage Session	Session Management	Team, Real-time, Data Access
UC-7: Manage Rounds	Session Management	Real-time, Data Access
UC-8: Submit Ideas	Idea Management	Session, Real-time, Data Access
UC-9: AI Assistance	AI Integration	Session, Data Access

	SOFTWARE DESIGN DOCUMENT for Brainstorming Application	Issue No: 1.0 Issue Date: Dec 2025
---	---	---

UC-10: Real-time Updates	Real-time Communication	Session
UC-11: Session Logs	Reporting & Logging	Session, Data Access
UC-12: Analytics	Reporting & Logging	Session, Data Access